

To address the above challenges, Google has released TensorNetwork, an open source library for efficient tensor calculations.

The Google AI team, which developed the library in collaboration with the



Perimeter Institute for Theoretical Physics, is hoping that it will become a valuable tool for physicists and machine learning practitioners in the future.

Google has introduced TensorNetwork in a series of papers, the first of which presents the new library and its API, and provides

an overview of tensor networks for a non-physics audience. The second paper focuses on a particular use case in physics, demonstrating the speed-up that one gets using GPUs.

TensorNetwork uses TensorFlow as a backend and is optimised for GPU processing, which enables speed-ups of up to 100x when compared to work on a CPU.

“We compare the use of CPUs with GPUs and observe significant computational speed-ups, up to a factor of 100, when using a GPU and the TensorNetwork library,” Google sources wrote in the company’s open source blog.

“TensorNetwork is a general-purpose library for tensor network algorithms, and so it should prove useful for physicists as well,” the blog post added.

In the future, Google is planning time series analysis on the ML side and quantum circuit simulation on the physics side. Together with the open source community, it will also continue to add new features to TensorNetwork.

ICFOSS to help women restart their careers in the free software domain

The International Centre for Free and Open Source Software (ICFOSS), an autonomous institution under the Kerala state government, is organising intensive

training sessions on various free software domains, for women who’ve taken career breaks.

Titled ‘Back to Work’, the initiative aims to empower a large number of ambitious and talented women, who remain

outside the workforce owing to marriage, motherhood, local constraints and family commitments, to restart their career in various free software domains.

The first training sessions under this initiative were on LaTeX and ‘Software testing’ and have been held at the Sports Hub in Thiruvananthapuram.

Graduates with prior experience in typesetting, scientific documentation and software testing (and have taken a career break) can apply for this training.

For the past few years, ICFOSS has been conducting various women empowerment programmes like ‘Women Hackathon’ and ‘Winter School for Women’ to build skills and confidence, helping this valuable human resource contribute more to technology.

For more news, visit www.opensourceforu.com



After 25 months of development, Debian 10 Buster finally arrives

The wait is over! The latest version of Debian Linux (code named Debian 10 Buster) has been released with numerous updated software packages.

The Debian Security team and the Debian Long Term Support team took more than two years (25 months) to develop the new version, which they say will be supported for the next five years.

Debian 10 Buster comes with improved security, better Unified Extensible Firmware Interface (UEFI) support as well as a new and improved CUPS printing system.

This newest release can run on most of the popular Linux desktops, including Cinnamon 3.8, GNOME 3.30, KDE Plasma 5.14, LXDE 0.99.2, LXQt 0.14, MATE 1.20, and Xfce 4.12.



By default, Debian uses the GNOME desktop. It relies on the Wayland display server instead of Xorg. However, the Xorg display server is still installed by default, and the default display manager allows users to choose Xorg as the display server for their next session.

In January this year, the Debian Desktop Team selected the theme ‘futurePrototype’ by designer Alex Makas as the default theme for Debian 10 Buster.

Debian 10 Buster also comes with numerous updated software packages, such as Apache 2.4.38, BIND DNS Server 9.11, Chromium 73.0, Emacs 26.1, Firefox 60.7 (in the firefox-esr package), GIMP 2.10.8, GNU Compiler Collection 7.4 and 8.3, GnuPG 2.2, Golang 1.11, etc.

Future Debian releases are expected to include tools and metadata to enable end-users to validate the provenance of packages within the archive.